

TABLEAU PROJECT DOCUMENTATION

Hospital Operations & Patient Analytics

984 patient records • Tableau • Hospital Operations Dashboard

Section	What this document covers
Project Overview	Purpose, scope and objectives of the Hospital Operations Dashboard.
Dataset	Source name, record count and main fields used in the Tableau workbook.
KPI Definitions	Meaning and purpose of the five main overview KPIs.
Dashboard Guide	Practical description of the dashboards and their visuals.
Methodology	How the data was prepared and converted into Tableau dashboards.
Observations & Conclusion	Main observations, limitations and final project summary.

1. Project Overview

This project is a Tableau-based Hospital Operations Dashboard created to understand patient activity and hospital performance from a patient-level dataset. The analysis turns raw patient information into practical visual reports covering patient volume, admission patterns, length of stay, billing, insurance, patient status, demographics and location.

2. Project Objectives

The project is designed to:

- Organize hospital patient data into a clear analytical view.
- Monitor the number of patients and important patient-related measures.
- Understand admission trends and different admission types.
- Analyse patient status, age group, gender, insurance and room type.
- Compare departments using patient count, length of stay and billing-related measures.
- Study patient distribution by city and state.
- Provide an interactive Tableau dashboard that is easy to read and explore.

3. Dataset Source and Description

Dataset source: Hospital_Data (Hospital_Dataset_984_Patients)

Record count: 984 patient records

No external dataset website is assumed in this documentation; the source is identified using the dataset name available in the Tableau workbook.

Age Group calculation: 0–18, 19–30, 31–40, 41–50, 51–60 and 60+.

Data preparation: The data was arranged for Tableau analysis using patient, date, categorical and numerical fields.

4. KPI Definitions

The KPI Overview worksheet contains five main measures that provide a quick summary before the detailed charts:

- Total patient count
- Average age
- Average length of stay
- Average total bill
- Average satisfaction rating

5. Dashboard Guide

Hospital Overview

Provides a high-level view of the hospital. Main patient KPIs are placed at the top, followed by patient distribution charts for quick understanding of the overall situation.

Patient Analytics

Focuses on patient details and patient movement, including age, admission type, admission trends, patient status and insurance type.

Department Analytics

Compares operational differences across departments and other hospital dimensions, including room usage, patient trends, length of stay, admission type by department and geographic distribution.

Resource Utilization

Combines location, time, billing, length of stay and hospital summary information to provide a broader operational view using the fields available in the dataset.

6. Healthcare Operations Methodology

Data Understanding: Reviewed the hospital patient data and identified fields suitable for patient, admission, operational, financial and demographic analysis.

Data Preparation: Prepared the data for consistent Tableau analysis using the hospital dataset containing 984 patient records.

Creating the Age Group: Grouped Age into 0–18, 19–30, 31–40, 41–50, 51–60 and 60+ categories.

KPI Creation: Selected the main overview measures: total patient count, average age, average length of stay, average total bill and average satisfaction rating.

Chart Development: Created Tableau worksheets for comparisons, trends, distributions, maps and summary views.

Dashboard Arrangement: Arranged the worksheets into four dashboards using KPI cards, charts and maps.

Interaction: Used gender-based filtering/action behaviour where available so viewers can explore relevant data points.

Validation: Reviewed KPI and chart values against the worksheet fields and checked that the dashboard layout was readable and logically grouped.

Overall workflow: Hospital Patient Data → Data Preparation → Calculated Age Groups → KPI Selection → Tableau Worksheets → Dashboards → Interactive Analysis

7. Key Observations

- The workbook contains 984 patient records in a patient-level dataset.
- The dashboard separates overall hospital information from detailed patient, department and operational views.
- Patient admissions are analysed over time using Admission Date at a monthly level.
- Patient status is separated into Deceased, Discharged, Recovered, Referred and Under Treatment.
- Age is converted into six practical groups for easier demographic comparison.
- Department-level views connect patient counts with admission type and length-of-stay measures.
- Billing is analysed together with patient activity and department-level measures.
- State and city maps provide a geographic view of patient distribution.

8. Limitations

- The analysis is limited to fields available in the supplied hospital dataset.
- The workbook is an analytical dashboard, not a live hospital information system.
- Measures such as bed occupancy or department efficiency should only be reported when the required capacity/efficiency data is available.
- The dashboard supports descriptive analysis and does not make clinical diagnoses or treatment recommendations.

9. Conclusion

The completed Hospital Operations Dashboard converts patient-level hospital data into connected analytical views. The first dashboard provides the overall picture, while the remaining views focus on patient details, department and operational dimensions, and resource/location-related measures. The final workbook provides a practical way to explore KPIs, charts, trends and maps instead of working directly with raw records.

End of Project Documentation